SAMPLE

Earth's Dynamic Systems: Wildfires

Fire on the Landscape

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Introduction

A wildfire is an unplanned, uncontrolled fire burning across vegetation, and it is one of the oldest forces life on Earth has ever dealt with: fire was shaping forests before the dinosaurs, and the plants of whole continents are built to burn. Yet the same phenomenon is a modern disaster, racing through suburbs, darkening cities a thousand kilometers downwind with smoke, and costing nations billions of dollars a year to fight. This book tells the whole story of wildfire, from the fossil charcoal of four hundred million years ago to the record seasons of the 2020s, in plain, unhurried language, with no equations and no prior training needed.

The chapters carry the science first: how a fire ignites and what the fire triangle really means, how wood actually burns, the types of fire and their speeds, the weather that drives them, and the climate that is slowly rewiring their seasons. Then the account widens to the consequences: the chemistry and reach of smoke, the toll in lives and money, the health of the people who breathe the aftermath, and the ecology of a force that is as creative as it is destructive, the maker of forests as much as their destroyer. The working chapters carry the human response: the towers, cameras, and satellites that watch for the first curl of smoke, the crews and aircraft that fight what follows, and the prevention, from prescribed burns to goat herds, that seeks to make the collision of people and fire survivable. A trivia chapter weighs the field's best stories honestly, and the closing chapters gather the great seasons of recent years, Black Summer in Australia and the record Canadian years, with the global picture the satellites now draw.

The honest thread through all of it is a paradox: fire cannot be abolished, because it is a working part of the living world, and it cannot be left alone, because people now live in its path. The book's argument, built chapter by chapter, is that the future of wildfire is not a war to be won but a relationship to be managed, learned by every generation that has tried the alternatives.

The charcoal record deserves a closing explanation, because it is a stranger archive than the reader may expect. Fossil charcoal, unlike ordinary coal, is burned tissue, and its microscopic structure survives so well that the wood's cell walls can still be identified; the fire-scar record of the living trees goes deeper still, because the big pines and redwoods heal over their burns, and a trunk cut or cored today shows its own fire history in dated scars, a century or more of burns written in rings. Put together, the charcoal in the soils and the scars in the trunks have let the fire historians reconstruct whole regimes, the frequency and season of the pre-human fires of the American West, the boreal, and the Mediterranean, and that reconstruction is the scientific foundation of the prescribed-fire policy the later chapters describe: the landscapes are managed toward measured historical fire regimes, not guessed ones.

The animal side of the deep history completes the chapter. The insects of the burns, the jewel and longhorn beetles that detect smoke and fly into it to lay eggs in the freshly dead wood, are among the most specialized of all animals, some with heat-sensing organs tuned to the temperatures of a recent fire; the birds follow, the black-backed woodpeckers and the flying predators of the open burns; and the mammals move through the mosaic of ages the fires maintain. Evolution has not merely tolerated fire; it has staffed the burned ground with a workforce, the chapter's closing image, and the modern reader should picture every forest fire as a scene with its own arriving specialists, a recruitment that begins within days.

A note on the continents' different fires belongs in the deep history, because the reader's mental picture depends on where they live. Australia is the fire continent, the eucalypts' oils and the nutrient-poor soils building a flora that burns enthusiastically and resprouts immediately, the aboriginal burning and the modern seasons two chapters of one long story; Africa is the fire's largest stage, the savannas' annual dry-season burning the planet's biggest burn by area; North America splits between the crown-fire conifers of the West and the surface-fire pines of the Southeast; Eurasia's taiga runs the lightning fires of the boreal; and Europe's Mediterranean, the smallest and most densely settled of the fire lands, carries the human-fire relationship at its tightest, every hectare of flammable hillside within sight of a village. The biosphere's fire regimes are as varied as its climates, the chapter's point, and every management doctrine the later chapters describe is a local dialect of one global grammar.

Chapter 1: Fire Before Us

The deepest history of wildfire is written in charcoal, and the record is astonishingly old. The oldest widely accepted fossil evidence of fire comes from the early land communities of the Silurian period, some four hundred twenty million years ago, where fossils of the giant fungi Prototaxites, organisms that stood meters tall when the

land was nearly bare of competition, have been found preserved as charcoal, burned in fires that predate the first forests. Fire, in other words, is older than trees; it followed life onto the land almost immediately, waiting only for enough oxygen in the atmosphere and enough fuel on the ground.

The oxygen history is the fire history, and the two rose together. In the Late Devonian, atmospheric oxygen stood around thirteen percent, enough for smoldering ground fires but little else; through the Carboniferous and into the Late Permian it climbed to thirty and thirty-one percent, well above today's twenty-one, and the ancient forests burned accordingly. The great coal-forming swamps of the Carboniferous, the Lepidodendron forests of club-trees, show charred peaks in their fossil layers, the preserved evidence of crown fires running through canopies tens of millions of years before any animal walked into one; at peak oxygen, even wet vegetation could carry fire, and the fossil record shows burning in swamps that no modern fire could cross.

From that deep beginning, fire became an evolutionary force, and the plants bear the scars and the tools. Whole lineages evolved to survive, resist, or exploit burning: the thick insulating bark of the pines and the redwoods, the serotinous cones that open only when a fire's heat melts their resin, the seeds that germinate in ash-enriched soil, the eucalypts of Australia that resprout from protected buds under their charred bark, and the grasses of the world's savannas that burn wholesale every dry season and regrow within weeks. The fire-prone habitats, in turn, shaped the dominance of some of the world's most successful trees, the pines of the north and the eucalypts of the south being the textbook cases, genera that prospered because fire killed their competitors and prepared their seedbeds. The reader should hold this chapter's picture for everything that follows: when a wildfire burns today, it is running a program the biosphere wrote hundreds of millions of years ago, and the modern question, which the later chapters take up, is not whether fire belongs but where people can safely live alongside a force this old.

A note on the modern policy vocabulary belongs here, because the reader will meet the terms in the news. Prescribed fire, controlled burning, and planned burning name the intentional fires set under management; wildland fire use names the natural ignitions allowed to burn for resource benefit; suppression names the extinguishing, full or partial; and the fire countries each keep their own additional dialect, Australia's backburning and hazard reduction, the Mediterranean's contrast between the fire-prevention culture of the village and the protection culture of the modern state. The vocabulary matters beyond the dictionary, because the century of suppression taught the agencies that the words on the page drive the money on the ground, and the modern reorganization of fire management, in the United States and elsewhere, has been in large part an effort to rewrite the doctrine so the words match the ecology, a bureaucratic struggle the ecologists have fought for fifty years and are still, by their own accounts, only winning.

The policy history's honest cost deserves its paragraph, because the reckoning is measurable. A century of suppression in the fire-adapted forests, in North America especially, left the woods dense with the fuels the frequent gentle fires once consumed, the modern name for the result being fuel buildup, and the fire behavior in the built-up fuels is worse than the historical fires in every measure, hotter, faster, and more lethal to the trees themselves; the forests that evolved with mild surface fires now burn as crown fires, killed wholesale by the same force they once shrugged off. The restoration of the historical fire, by prescribed burning and by managed wildfire, is the task the agencies now describe as the fire debt, a backlog measured in millions of hectares, and the seasons of recent decades, in the western United States above all, are the interest the debt accrues by, a metaphor the fire services themselves use, and worth the reader's keeping.

The technological turn in the human story deserves its paragraph, because the twentieth century changed the fire's handlers, not just its handling. The aircraft, the radios, the engines, and the lookouts gave the agencies a reach their predecessors could not imagine, and the aerial century's statistics, fires caught at sizes the previous era could not have touched, are the suppression era's real achievements, worth recording beside its failures; the same century moved the people, the suburbs and the exurbs spreading into the fuels, the recreation economy carrying ignition into the backcountry, and the power grid stringing ignition across the windiest ridges. The modern fire problem, the chapter's closing observation, is the collision of these two arcs, the technology that can fight almost any fire meeting the development that keeps building things to burn, and the reader who wonders why the richest era's fire losses keep rising should hold this paragraph: the fires are not the variable that got away, the exposure did.